Scalability & Future Plan

Date	30 September 2026
Team ID	3
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models

This document explains how ComicCraft can grow to serve more users and how it will evolve.

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Comic generation runs inside the web request and image inference is serialised	Users wait minutes; concurrent users queue for the GPU	High
2	No login, rate limiting or content moderation	Anyone can consume API quota; risk of unsuitable prompts	High
3	Generated files stay on local disk with no clean-up and no database	Disk grows over time; comics cannot be revisited later	Medium
4	A single Stable Diffusion model per process; no character consistency across panels	Limited visual continuity	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	One Uvicorn process, work done in request threads	Background job queue (Celery/RQ + Redis) with progress polling; several workers behind Nginx or Cloud Run
2	Data Storage	Local folders static/panels and static/exports	Cloud object storage for images and PDFs, database for users and comics, scheduled clean-up
3	Performance	Sequential images, 25 inference steps, GPU or CPU	GPU inference service, faster/distilled image model, parallel panels across GPUs, caching of outlines
4	Security	Secrets in .env, input validation	User authentication, rate limiting, prompt/content moderation, secret manager, HTTPS

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Accounts, saved comic library, regenerate a single panel, edit text	Q4 2026	Users can revisit and refine comics
Phase 3	Consistent characters, voice narration, multi-language stories, content-safety filter	Q1 2027	Higher quality and wider audience
Phase 4	Cloud deployment with GPU, CI/CD, share links, more panel counts and layouts	Q2 2027	Reliable public service that scales